

5. Agriculture Crop Yield Dataset

Question 5

An agricultural department wants to analyze crop yield based on rainfall, temperature, fertilizer usage, soil type, and crop yield. Perform EDA to identify the factors influencing crop production.

Farm_ID	Rainfall_mm	Temperature	Fertilizer_kg	Soil_Type	Crop_Yield_kg
F01	850	28	60	Loamy	3200
F02	600	32	45	Sandy	2100
F03	900	27	65	Loamy	3500
F04	500	34	40	Sandy	1800
F05	780	29	55	Clay	3000
F06	650	31	48	Sandy	2300
F07	920	26	70	Loamy	3700
F08	720	30	52	Clay	2800
F09	550	33	42	Sandy	1900
F10	800	28	58	Loamy	3100
F11	880	27	62	Loamy	3400
F12	620	32	46	Sandy	2200
F13	760	29	54	Clay	2950
F14	940	26	72	Loamy	3800
F15	580	33	44	Sandy	2000
F16	820	28	60	Clay	3150
F17	700	30	50	Clay	2700
F18	960	25	75	Loamy	3900
F19	540	34	41	Sandy	1850
F20	790	29	56	Clay	3050

Common EDA Tasks:

1. Identify the number of rows and columns.
2. Check data types of each column.
3. Find missing values and duplicate values.
4. Calculate mean, median, minimum, maximum, and standard deviation.
5. Draw suitable charts such as bar chart, histogram, scatter plot, and box plot.
6. Identify important patterns from the dataset.
7. Write 4 to 5 observations based on EDA.
8. Give a short conclusion from the analysis.

Answer

```
# =====  
# Agriculture Crop Yield EDA  
# =====
```

```
import pandas as pd  
import matplotlib.pyplot as plt
```

```
# Load Dataset  
df = pd.read_csv("agriculture_crop_yield_dataset.csv")
```

```

# =====
# 1. Number of Rows and Columns
# =====
print("Shape of Dataset:", df.shape)
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])

# =====
# 2. Data Types
# =====
print("\nData Types")
print(df.dtypes)

# =====
# 3. Missing and Duplicate Values
# =====
print("\nMissing Values")
print(df.isnull().sum())

print("\nDuplicate Values:", df.duplicated().sum())

# =====
# 4. Statistical Summary
# =====
print("\nMean")
print(df.mean(numeric_only=True))

print("\nMedian")
print(df.median(numeric_only=True))

print("\nMinimum")
print(df.min(numeric_only=True))

print("\nMaximum")
print(df.max(numeric_only=True))

print("\nStandard Deviation")
print(df.std(numeric_only=True))

# =====
# Chart 1 : Crop Yield by Soil Type (Bar Chart)
# =====
plt.figure(figsize=(7,5))

df.groupby("Soil_Type")["Crop_Yield_kg"].mean().plot(
    kind="bar",
    color=["green", "orange", "skyblue"]
)

plt.title("Average Crop Yield by Soil Type")

```

```

plt.xlabel("Soil Type")
plt.ylabel("Average Crop Yield (kg)")
plt.xticks(rotation=0)
plt.grid(axis='y', linestyle='--')
plt.show()

# =====
# Chart 2 : Rainfall Distribution (Histogram)
# =====

plt.figure(figsize=(7,5))

plt.hist(df["Rainfall_mm"], bins=6, color="lightgreen", edgecolor="black")

plt.title("Rainfall Distribution")
plt.xlabel("Rainfall (mm)")
plt.ylabel("Frequency")
plt.grid(True)
plt.show()

# =====
# Chart 3 : Rainfall vs Crop Yield (Scatter Plot)
# =====

plt.figure(figsize=(7,5))

plt.scatter(df["Rainfall_mm"], df["Crop_Yield_kg"],
            color="blue", s=90)

plt.title("Rainfall vs Crop Yield")
plt.xlabel("Rainfall (mm)")
plt.ylabel("Crop Yield (kg)")
plt.grid(True)
plt.show()

# =====
# Chart 4 : Crop Yield Box Plot
# =====

plt.figure(figsize=(6,5))

plt.boxplot(df["Crop_Yield_kg"])

plt.title("Crop Yield Box Plot")
plt.ylabel("Crop Yield (kg)")
plt.grid(True)
plt.show()

# =====
# Correlation
# =====

print("\nCorrelation Matrix")
print(df[["Rainfall_mm", "Temperature", "Fertilizer_kg", "Crop_Yield_kg"]].corr())

```

```
# =====
# 6. Important Patterns
# =====
print("\n===== Important Patterns =====")
print("1. Loamy soil gives the highest average crop yield.")
print("2. Higher rainfall generally results in higher crop production.")
print("3. Higher fertilizer usage is associated with higher crop yield.")
print("4. Extremely high temperatures are associated with lower crop yield.")

# =====
# 7. Observations
# =====
print("\n===== Observations =====")
print("1. Loamy soil farms produce the highest yield.")
print("2. Sandy soil farms produce the lowest yield.")
print("3. Rainfall and fertilizer usage positively influence crop yield.")
print("4. Crop yield decreases when temperature becomes too high.")
print("5. No missing values or duplicate records are found.")

# =====
# 8. Conclusion
# =====
print("\n===== Conclusion =====")
print("The dataset shows that rainfall, fertilizer usage, and soil type significantly influence crop production. Loamy soil with adequate rainfall and proper fertilizer application achieves the highest crop yield. The dataset is clean and suitable for further predictive analysis.")
```

Output:

Shape of Dataset: (20, 6)

Rows: 20

Columns: 6

Data Types

```
Farm_ID      object
Rainfall_mm  int64
Temperature   int64
Fertilizer_kg int64
Soil_Type     object
Crop_Yield_kg int64
dtype: object
```

Missing Values

```
Farm_ID      0
Rainfall_mm  0
Temperature   0
Fertilizer_kg 0
Soil_Type     0
Crop_Yield_kg 0
dtype: int64
```

Duplicate Values: 0

Mean

Rainfall_mm	743.00
Temperature	29.55
Fertilizer_kg	54.75
Crop_Yield_kg	2820.00

dtype: float64

Median

Rainfall_mm	770.0
Temperature	29.0
Fertilizer_kg	54.5
Crop_Yield_kg	2975.0

dtype: float64

Minimum

Rainfall_mm	500
Temperature	25
Fertilizer_kg	40
Crop_Yield_kg	1800

dtype: int64

Maximum

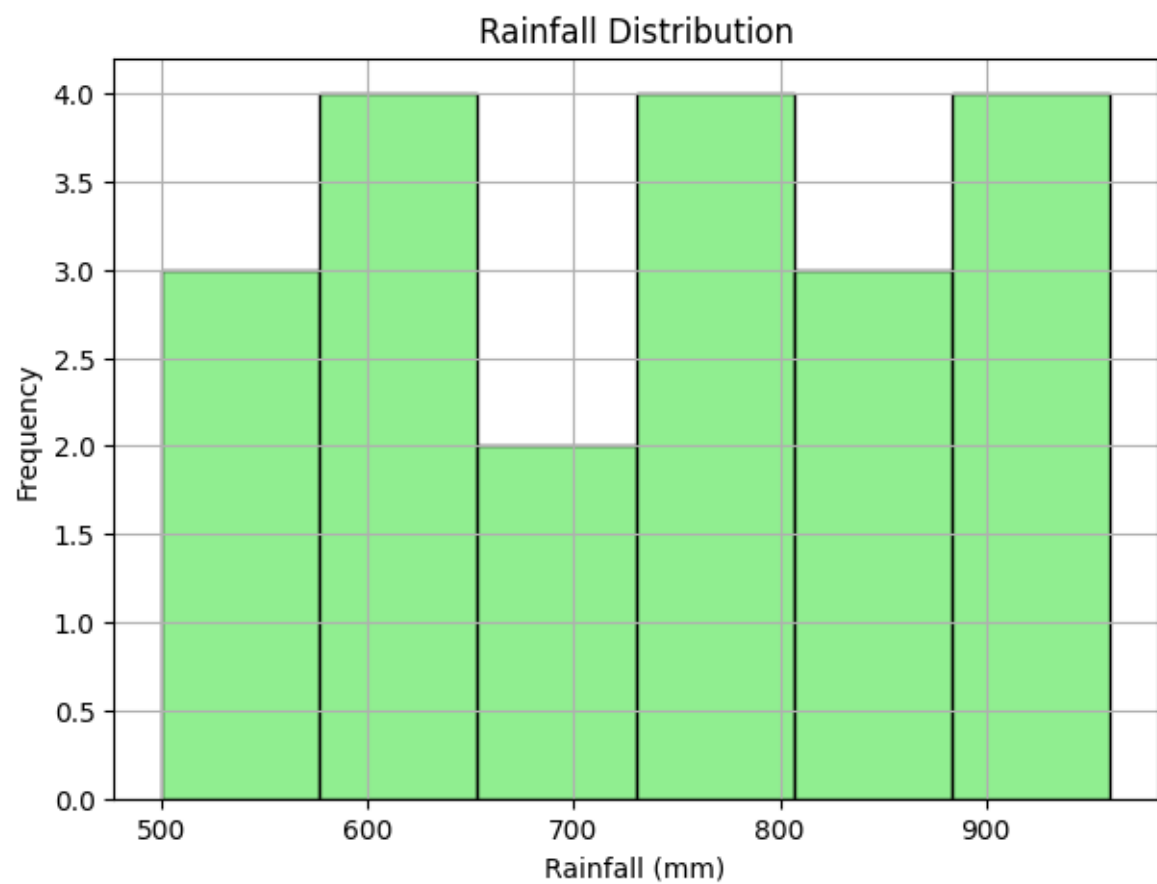
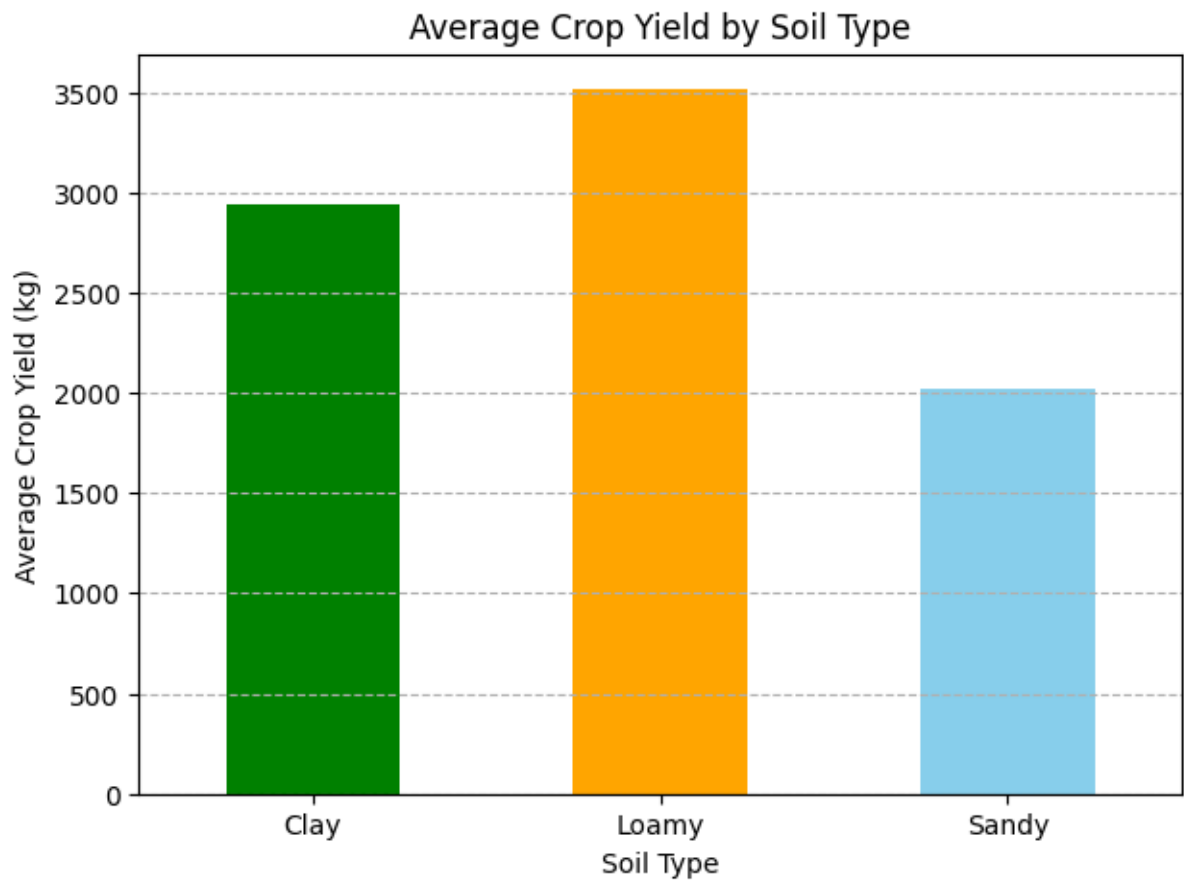
Rainfall_mm	960
Temperature	34
Fertilizer_kg	75
Crop_Yield_kg	3900

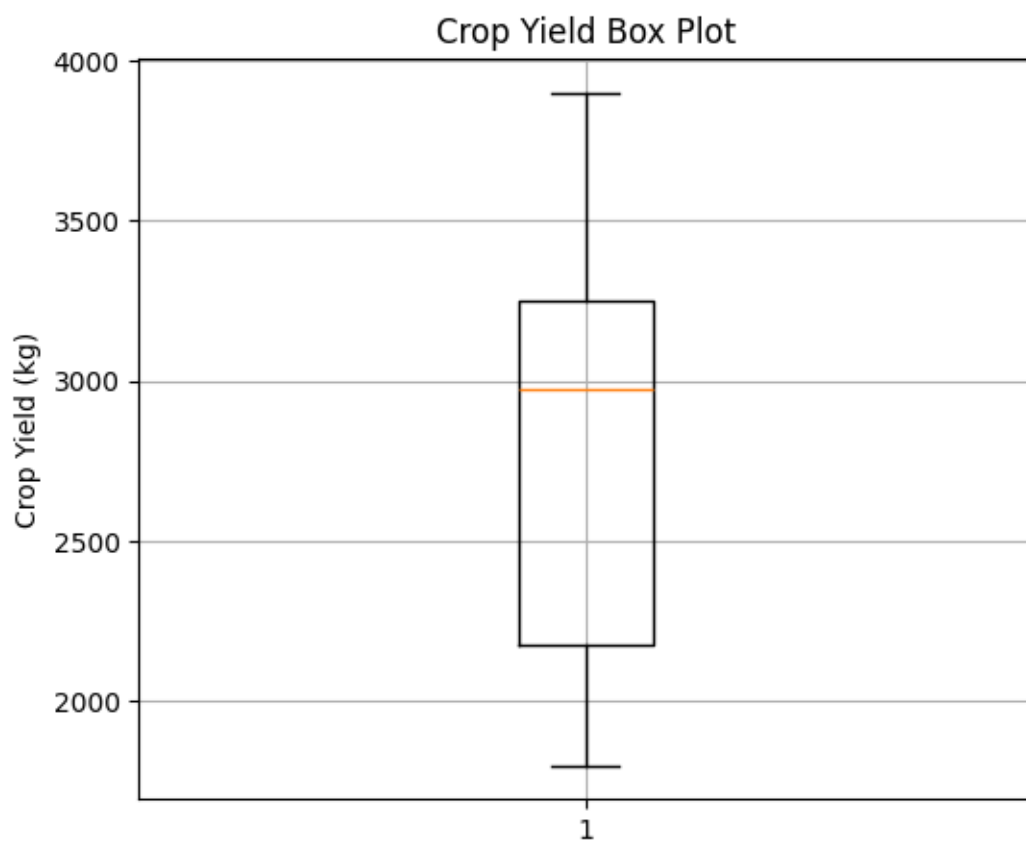
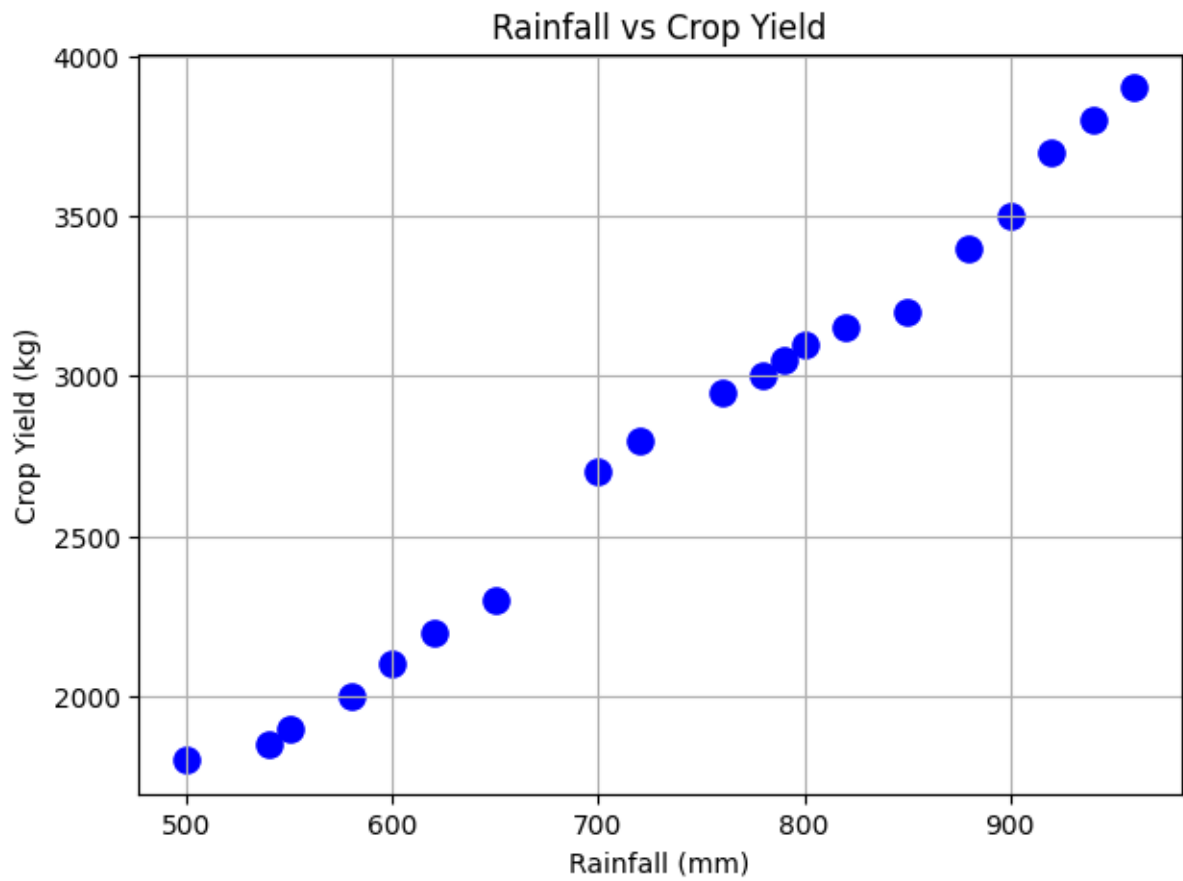
dtype: int64

Standard Deviation

Rainfall_mm	144.335136
Temperature	2.762055
Fertilizer_kg	10.507516
Crop_Yield_kg	681.407213

dtype: float64





Correlation Matrix

Rainfall_mm Temperature Fertilizer_kg Crop_Yield_kg

Rainfall_mm	1.000000	-0.993192	0.980550	0.995521
Temperature	-0.993192	1.000000	-0.979735	-0.993300
Fertilizer_kg	0.980550	-0.979735	1.000000	0.983181
Crop_Yield_kg	0.995521	-0.993300	0.983181	1.000000

===== Important Patterns =====

1. Loamy soil gives the highest average crop yield.
2. Higher rainfall generally results in higher crop production.
3. Higher fertilizer usage is associated with higher crop yield.
4. Extremely high temperatures are associated with lower crop yield.

===== Observations =====

1. Loamy soil farms produce the highest yield.
2. Sandy soil farms produce the lowest yield.
3. Rainfall and fertilizer usage positively influence crop yield.
4. Crop yield decreases when temperature becomes too high.
5. No missing values or duplicate records are found.

===== Conclusion =====

The dataset shows that rainfall, fertilizer usage, and soil type significantly influence crop production. Loamy soil with adequate rainfall and proper fertilizer application achieves the highest crop yield. The dataset is clean and suitable for further predictive analysis.